

Chapter Two

The Research Process

Introduction

Before embarking up on the details of research methodology and techniques, it seems appropriate to present a brief overview of the research process. The research process consists of a number of closely related activities. These activities/steps overlap continuously rather than following a strictly prescribed sequence. They do not necessarily follow each other in any specific order. However, the following order provides a useful procedural guideline regarding the research process.

1. Formulating the Research Problem:

The first step in the research process is the choice of a suitable problem for investigation. Research starts with a felt difficulty. It takes place when there is a problematic situation and a need to solve the problem. Sources of a research problem are one's professional experience, theory, and professional literature. Initially the problem may be stated in a broad general way and then narrowed down to a specific research problem. That means, after understanding the problem thoroughly the same has to be rephrased in to meaningful terms from analytical point of view.

A proper definition of a research problem will enable the researcher to be on the track whereas an ill-defined problem may create hurdles.

Thus, defining a research problem properly is a prerequisite for any study and is a step of the highest importance. In fact, formulation of a problem is often more essential than its solution.

The Components of the Research Problem

A research problem is one which requires a researcher to find out the best solution for the given problem, i.e., to find out by which course of action the objective can be attained optimally in the context of a given environment. Hence, in a given research problem

- ✓ There must be an individual or a group which has some difficulty or problem.
- ✓ There must be some objective(s) to be attained. If one wants nothing, one

cannot have a problem

- ✓ There must be alternative means (or courses of action) for obtaining the objective(s) one wishes to attain. If a researcher has no choice of means, he can not have a problem.
- ✓ There must remain some doubt in the mind of a researcher with regard to the selection of alternatives. This means, the research must answer the question concerning the relative efficiency of the possible alternatives.
- ✓ There must be some environment(s) to which the difficulty pertains.

The research problem should be defined in a systematic manner, giving due attention to all relating points. The technique for this purpose involves the following general steps. These are:

- (i) Statement of the problem in a general way (choice of research topic)
- (ii) Understanding the nature of the problem;
- (iii) Surveying the available literature;
- (iv) Developing ideas through discussions; and
- (v) Rephrasing the research problem into a working proposition

(i) Statement of the Problem in a General Way: First of all, the problem should be stated in a broad general way, keeping in view either some practical concern or some scientific or intellectual interest. It is sometimes advisable to do some sort of observation and preliminary survey or what is often called pilot survey. Then the researcher can himself state the problem or he can seek the guidance of the guide or the subject expert in accomplishing this task.

(ii) Understanding the Nature of the Problem: The next step in defining the problem is to understand its origin and nature clearly. The best way of understanding the problem is to discuss it with those who first raised it in order to find out how the problem originally came about and with what objectives in view. If the researcher has stated the problem himself, he should consider once again all those points that induced him to make a general statement concerning the problem. He can enter into discussion with those who have a good knowledge of the problem concerned or similar other problems. The researcher should also keep in view the environment with in which the problem is to be studied and understood.

(iii) Surveying the Available Literature: All available literature concerning the problem at hand must necessarily be surveyed and examined before a definition of the research problem is given. This is done to find out what data and other materials, if any, are available for operational purposes. Surveying literature would also help a researcher to know if there are certain gaps in the theories, or whether the existing theories applicable to the problem under study are inconsistent with each other, or whether the findings of the different studies do not follow a pattern consistent with the theoretical expectations and so on. Studies on related problems are useful for indicating the type of difficulties that may be encountered in the present study as also the possible analytical shortcomings. At times such studies may also suggest useful and even new lines to the present problem.

(iv) Developing the Ideas through Discussions: Discussions concerning a problem often produces useful information. Various new ideas can be developed through such an exercise. Hence, a researcher must discuss his problem with his colleagues and others who have enough experience in the same area or in working on similar problems. This is known as an experience survey. Discussions with experienced persons should not only be confined to the formulation of the specific problem at hand, but should also be concerned with the general approach to the given problem, techniques that might be used, possible solutions, etc.

(v) Rephrasing the Research Problem: Finally, the researcher must rephrase the research problem in to a working proposition. Rephrasing the problem into analytical or operational terms is not a difficult task. Through rephrasing, the researcher puts the research problem in as specific terms as possible so that it may become operationally viable and may help in the development of working hypotheses.

In rephrasing research problem, a researcher has to narrow a research focus, and develop, *the statement of the problem*

A good problem statement should be clearly written and be able to immediately capture the readers. It should be able to advance knowledge or be of interest to society. In addition, it should indicate the scope of the study and be objectively researchable. Finally, the problem statement must give the purpose of the research (Mugenda and Mugenda, 1999). A good research problem should be capable of being stated in one major research question, which reveals the research gap. Such

question(s) may be descriptive, relationship questions, or difference questions.

The problem statement conveys the specific research problem, usually in a few paragraphs. It should indicate:

- ✓ The focus area of the research (the issue raised)
- ✓ Indicates researches done by other researchers in the area including their findings
- ✓ Indicate issues that are not yet assessed by other researcher in the area (the knowledge gap or unsolved questions which appears to justify another research.
- ✓ Show difference in the findings of various researchers in the area (if any)
- ✓ Indicate the study area and the issues or subjects of research
- ✓ Indicate the purpose of the study
- ✓ Indicate what is meant to be shown or done by the research

Research Objective- While statement of purpose helps to focus the study, research objectives further specify the issues or area of knowledge that the researcher wants to examine or explore. Stating research objectives entails breaking down the scope of the stated purpose to enable in-depth investigation. Usually, there is one general research objectives, while the number of specific research objectives varies from one research to another, but their number should be reasonable enough to properly address by a particular research.

Suppose that our title says: “An Analysis of the Impact of Training on the Workers’ Efficiency at Work’: the case of XY- Company”. In line with the above research topic one can develop research objectives as follows:

The overall purpose of this research is to analyze the impact of training on Efficiency of Workers at Work. More specifically, it aims at addressing the following specific objectives

- i) To determine if training affect workers’ speed at work
- ii) To investigate the impact of training on workers ability of discharging their responsibility properly
- iii) To examine if training affects workers respect for their work
- iv) To describe the relationship between workers’ access to training and their

productivity

Research Questions- Research questions constitute the most important elements of any research. They describe the ideas contained in a research objective. However, research questions emerge after the research objectives. They point out the data that are required to be collected in a study. According to Blaikie (2000), the questions have to be related to three aspects: What, Why and How? The ‘What’ questions seek descriptions, ‘Why’ questions seek explanation and understanding, and the ‘how’ questions seek interventions to bring about change? In relation with the above research objectives, one may formulate the following research questions.

- i) What is the effect of training on workers’ speed at work?
- ii) What is the impact of training on the workers ability of discharging their ability?
- iii) Is there any difference in workers respect for their job because of access or non-access to training?
- iv) What relationship is there between workers’ productivity and access to training opportunity?

2. Review the Literature:

Once the problem is formulated, a brief summary of it should be written down. At this juncture, the researcher should undertake extensive literature survey connected with the problem. The researcher may review two types of literature-conceptual and empirical. Conceptual literature is concerned with related concepts and theories. Empirical literature is about studies made earlier which are similar to the one proposed.

3. Formulation of Working Hypothesis:

After extensive literature survey, researcher should state in clear terms the working hypothesis or hypotheses. Hypothesis is tentative assumption made in order to draw out and test its logical or empirical consequences. Hypotheses provide focal point for research. The role of hypothesis is to guide the researcher by delimiting the area of research and to keep him on the right truck. However, we may encounter a problem where we do not need working hypotheses.

A research hypothesis is a tentative proposition or prediction of the expected relationships between two or more variables. This tentative proposition(s) is/are subject to verification upon the undertaking of the research work. If the finding of the research supports our earlier predication, then the hypothesis will be accepted, otherwise it will be rejected. A hypothesis is a tentative statement or an educated guess suggesting possible relationship, difference, or case between two or more variables that can be investigated and tested.

4. Preparing the Research Design (Including Sample Design):

The research problem having been formulated in clear cut terms, the researcher will be required to prepare a research design. This is stating the conceptual structures with in which research would be conducted. In addition, the researcher must design the way of selecting a sample or what is known as the sample design.

5. Types and Source of Data

There are two types of data which is usually used in researches, primary and secondary data. Primary data does not actually exist until and unless it is generated through the research process as part of the consultancy or dissertation or project. It would often be collected through techniques such as experimentation, interviewing, observation and surveys. On the other hand Secondary sources are information sources that, describe, or draw conclusions based on works written by others. In addition to this, Secondary sources are often referred to as being “one step removed” from the actual occurrence or fact. Again secondary source consists of a review of relevant literature conducted to assess the previous research studies conducted by other researchers and some statistical reports related to the topic of the study were reviewed. In addition review of various books, journals, articles, and notes from the internet pertaining to the study.

6. Sample design and sampling techniques

Kothari (2004) states that a sample design is a definite plan for obtaining a sample from a given target population and it refers to the techniques or procedures the researcher would adopt in selecting items for sample. According to Saunders et. al. (2009) unlike probability sampling, there are no rules in non-probability sampling techniques. Non-probability sampling involves the selection of subject based on

assumption regarding the population of interest, which forms the criteria for selection.

Example: If the population is infinite or unknown, how can we calculate the sample size?.

Cochran (1977) developed a formula to calculate a representative sample for infinite populations.

$$n_o = \frac{Z^2 pq}{e^2}$$

$$n_o = 1.96^2 * 0.5 * 0.5 / 0.05^2 = 385$$

Where, n_o is the sample size, z is the selected critical value of desired confidence Level, the value for Z is found in statistical tables which contain the area under the normal curve $Z = 1.96$ for 95 % level of confidence, p is the estimated proportion of an attribute that is present in the population, which is 0.5 and $q = 1 - p$ and e is the desired level of precision.

7. Collecting the Data (Execution):

The researcher should select one of the methods of data collection taking in to consideration the nature of investigation, objective and scope of the inquiry, financial resources, available time, and the desired degree of accuracy. Then using the selected method, adequate and dependable data have to be collected.

8. Analysis of Data (Including Test of Hypothesis):

After the data have been collected, the researcher turns to the task of analyzing them. The analytical techniques have to be appropriate and adequate. After analyzing the data, the researcher is in apposition to test the hypothesis, if any, he had formulated earlier.

9. Drawing Conclusions:

This is making generalizations after the analyses have been made. The conclusions must be based on the data and the analysis within the framework of the research study. Finally, the researcher has to prepare the report of what has been done by him.

